Energy 229 (2021) 120777

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Imputation of missing data from offshore wind farms using spatio-temporal correlation and feature correlation

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ARTICLE INFO

Article history:
Received 8 December 2020
Received in revised form
23 April 2021
Accepted 26 April 2021
Available online 30 April 2021

Keywords:
Offshore wind farm
Missing data imputation
Spatio-temporal correlation
Feature correlation

ABSTRACT

Many industrial applications, such as fault diagnosis and remaining useful life prediction, require high-dimensional inputs to predict a reliable output. For offshore wind farm supervisory control and data acquisition (SCADA) systems, unfortunately, signal inputs are often missing due to harsh weather, resulting in the failure of network/sensors. It limits the accuracy of subsequent diagnostic or prognostic tasks. Although many methods have been proposed for imputing missing data, their applicability in offshore wind farms is still problematic because wind turbines (WTs) are time-varying systems, and conventional learning methods require high computational cost. To address this problem, we propose a learning framework containing two learning models, corresponding to two missing-data conditions. The framework imputes missing data by designing a spatio-temporal correlation method for entire feature-missing conditions and a feature-correlation method for partial feature-missing conditions, respectively. A real-world offshore wind farm dataset of a SCADA system with 33 WTs and 68 features, which was recorded over a one-month period, is used for experimental validation. We demonstrate that the proposed framework imputes the missing data with much smaller mean absolute error (MAE) and mean squared error (MSE) and requires less computational time, compared to the existing machine-learning methods for both imputation conditions.

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1. Introduction

Wind power, as a low-carbon energy source, is one of the rapid-growing technologies for electricity production worldwide [1]. In environmental terms, it is a green strategy to achieve economic benefit and competitiveness [2]. Furthermore, WTs generate power without any emissions, and wind power will never be depleted, unlike conventional energy sources such as fossil fuels, coal, and natural gas. In general, wind-power energy can be grouped into two categories according to the location of the WTs, i.e., onshore or offshore wind farms [3]. At present, wind-power energy is mostly produced by onshore wind farms. However, their expansion is limited by the availability of land around large cities and the low wind speeds on land [4].

For large-scale electricity generation over the past few years, offshore wind farms have attracted more investment than onshore wind farms, as they require less space and use the higher wind speed of the open sea [5,6]. Unfortunately, complete data of WTs from SCADA systems is difficult to obtain due to severe adverse weather conditions in offshore areas, which cause network congestion or sensor failure, resulting in different degrees of data loss, causing errors in data processing and decreasing statistical efficiency [7,8]. In addressing this issue, it is not feasible to acquire all the datasets from real wind farms. Instead, researchers and engineers often generate virtual data to replace the missing data [9]. This procedure is referred to as data imputation. As soon as the missing values have been imputed, the complete dataset can be applied to diagnostic and prognostic tasks, such as condition monitoring, fault diagnosis, power generation, and prediction of remaining useful life of WTs [10–14]. Previous studies [7,14,15] have proved that the performance of power prediction and fatigue-assessment tasks of wind farms can be improved by the implementation of proper data-imputation methods. Hence, the imputation of missing values is of great significance to WT studies [16].

There are several standard machine-learning methods to impute missing data, such as decision trees, Gaussian process,

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https://doi.org/10.1016/j.energy.2021.120777
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Nomenclature

ν	the number of extended points
ω	the weight of the shift number
Φ	the number of segments
$\rho(\cdot)$	the penalty value of the shift number
$\mathbf{b}^p$	the bias factor between $\tilde{\mathbf{X}}_e^p$ and $Y_e^{n,p}$
$\mathbf{C}^p$	the cross-correlation coefficient matrix of the p_{th} feature
K	the number of correlated WTs
l	the max number of shift
N	the number of WTs
P	the number of features
Q_1	the first quartile
Q_3	the third quartile
r_φ	the boundary of the φ_{th} segment
S	the number of WTs with the smallest MAEs
T	the number of sample points
$X^{n,p}$	the observed time series of the p_{th} feature of the n_{th} WT
$Y^{n,p}$	the time series to be imputed of the p_{th} feature of the n_{th} WT
CQV	Coefficient of Quartile Variation
GNN	Graph Neural Network
kNN	k-Nearest Neighbors
KR	Kernel Regression
LASSO	Least Absolute Shrinkage and Selection Operator
LR	Linear Regression
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MSE	Mean Squared Error
PCHIP	Piecewise Cubic Hermite Interpolating Polynomial
RF	Random Forest
SCADA	Supervisory Control and Data Acquisition
SVM	Support Vector Machine
WT	Wind Turbine

Gaussian mixture models, etc [7,14,17]. Recently, neural network models (e.g., multilayer perceptron [15], recurrent neural networks [18], generative adversarial networks [19]) have been explored to tackle the imputation difficulties in terms of large-scale and multivariable time series with missing values. Most of these methods are based on an assumption of WTs: patterns of models for the training data and test data are identical, meaning that dataset distributions in training and test sets are unchanged over time. However, in practice, with the aging of WTs and the environmental change, the function relationship across features will change slightly; namely, it is a time-varying system. A fixed model is not optimal to fit all data patterns, and it is hard to perform high imputation precision at all times. Therefore, online-learning algorithms are necessary.

Another technical issue existing in wind farm data imputation is that missing-data conditions are different. Fig. 1 generally depicts two missing-data conditions of offshore wind farms. Fig. 1(a) refers to the condition that all features are missing at one of the WTs over a period of time, on the other hand, Fig. 1(b) shows the second condition that one of the features is missing at all or most of the WTs. To address these problems, an ensemble-learning method tackling both conditions within one framework has to be developed.

In this study, we propose an online-ensemble framework that consists of two learning models, each model able to deal with one missing-data condition. The proposed framework imputes missing data by designing a spatio-temporal correlation method (named "Lazy ST algorithm") for entire feature-missing conditions and an inter-feature correlation method (named "Online Variation algorithm") for partial features-missing conditions, respectively. Our framework can be applied to real-world offshore wind farms where missing data occurs frequently and randomly. We show that the proposed framework imputes missing values of WTs with much better performance in terms of imputation accuracy and computational time.

The proposed framework has several advantages. First, it can be referred to as an online-learning algorithm that estimates the missing data depending on its neighboring data. Therefore, the imputation is more accurate for time-variant WT systems. It is also well-suited to data acquired from sensors with high sampling frequencies, such as WT SCADA systems, saving much computational effort compared to other machine-learning algorithms. When

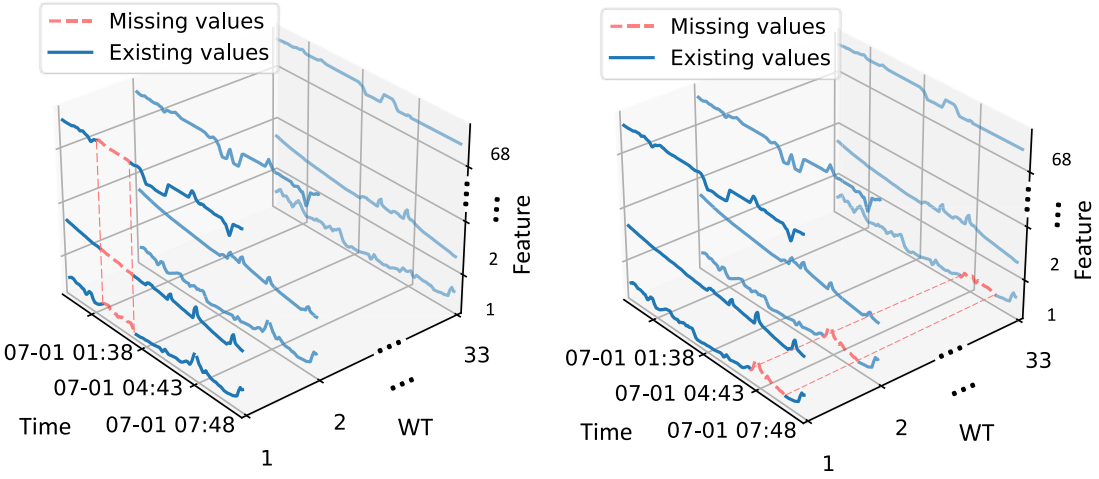


Fig. 1. Time-series plots of WTs with different features. There exists two missing conditions: (a) all features are missing at one of the WTs, and (b) one of the features at all or most of the WTs are missing, over a period of time.

dealing with extreme randomness and uncontrollable wind scenarios, the lazy-learning algorithm in this study achieves good results in terms of high precision, efficiency, and low-parameter dependence. Finally, this approach can be extended to other similar tasks for spatio-temporal missing-data imputations.

The rest of this paper is organized as follows. Section 2 introduces the algorithm framework. Section 3 presents experiment results and discussion. The conclusion is summarized in Section 4.

2. Related algorithms

2.1. Framework

The proposed ensemble framework for the data imputation of wind farm data is shown in Fig. 2. It involves two parallel algorithms for different missing-data conditions:

1. The first algorithm will be described in Section 2.2.1. Since a flat homogeneous ocean surface has less obstacles compared to a land surface, signal correlations between offshore WTs should be much higher than those of the onshore's. Even different WTs are aligned, with the wind velocity deficit in wake regions [20], their energy intensities vary over time. According to these characteristics of wind energy, when one of the WTs fails for a period of time as Fig. 1(a), a lazy-learning algorithm is designed to impute the missing values by learning the spatio-temporal correlations between the failed WT and other functional WTs (see in Algorithm 1).
2. The second algorithm will be described in Section 2.2.2. When one feature of all the WTs is missing for a short period of time as Fig. 1(b), an online-learning algorithm is proposed to fit the data patterns in a real-time manner (see in Algorithm 2). This algorithm is able to deal with a time-varying system.

2.2. Ensemble-learning framework

Consider that the offshore wind farm has N WTs, where each WT has acquired T sample points in total and each sample has P features.

2.2.1. Spatio-temporal correlation lazy-learning algorithm

In terms of the missing condition (a) in Fig. 1, when all features of a specific WT are missing over a period of time, missing values

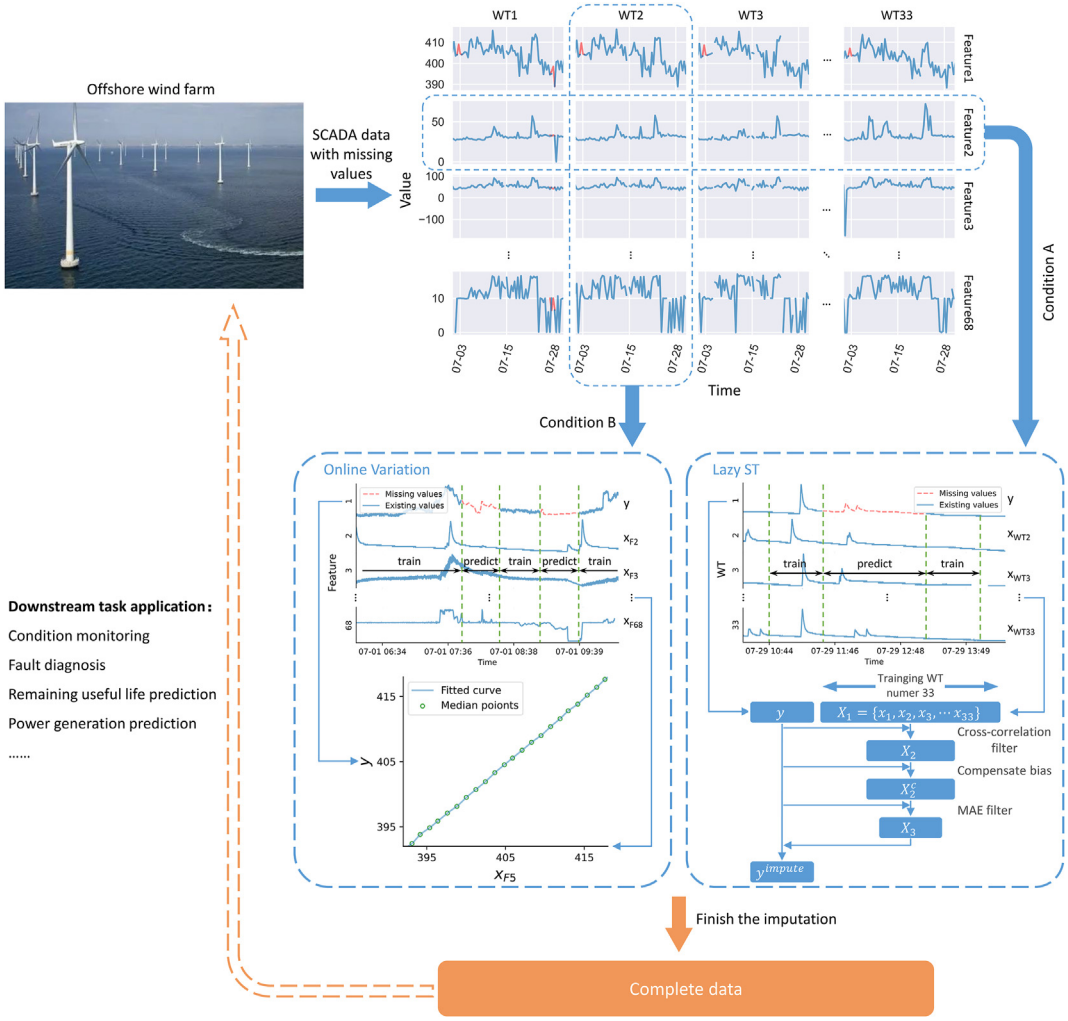


Fig. 2. Missing data imputation framework for offshore wind farms. The framework involves two learning algorithms: (1) a lazy learning algorithm is designed to impute the missing values by learning the spatio-temporal correlations between the failure WT and other functional WTs, and (2) an online variation method is designed to impute the partial feature missing case by using XGBoost method for feature selection and piecewise cubic Hermite interpolating polynomial for data imputation respectively. After all missing data is imputed, the complete dataset can be applied on different health monitoring and prediction tasks.

within this time period are not highly related to time series outside this period due to the randomness of wind power. To address this problem, the spatio-temporal correlation relationship between this failed WT and other functional WTs should be considered.

We impute the missing features from feature $p = 1$ to P. The object is to design a lazy-learning framework (denoted by "Lazy ST") that learns to impute missing values of p_{th} feature of the n_{th} WT on the basis of features' temporal as well as spatial correlations between the n_{th} WT and other $N-1$ well-aligned WTs.

Assume t_0 is the timestamp at the beginning of the missing segment, and t_1 is the timestamp at the end, with $1 < t_0 < t_1 < T$. Define the missing dataset $Y^{n,p}$ of p_{th} feature of the n_{th} WT as a sequence of time-series data as:

$$Y^{n,p} = \left\{x_i^{n,p}\right\}_{i=t_0}^{t_1} \in \mathbb{R}^{1\times(t_1-t_0+1)}, \tag{1}$$

where $p = 1, 2, \ldots, P$ and $n \le N$. The rest $N-1$ WTs construct the observed dataset, which can be defined as:

$$\mathbf{X}^p = \left[X^{1,p}, \ldots, X^{n-1,p}, X^{n+1,p}, \ldots, X^{N,p}\right]^T \in \mathbb{R}^{(N-1)\times(t_1-t_0+1)}, \tag{2}$$

where each element of $\mathbf{X}^p$ represents the observed data points of that WT. For example, for the $n-1_{\text{th}}$ WT, its observed data points from t_0 to t_1 are $X^{n-1,p} = \{x_i^{n-1,p}\}_{i=t_0}^{t_1}$.

To make use of spatio-temporal correlation relationship, we extend the observed dataset $\mathbf{X}^p$ and the missing dataset $Y^{n,p}$ by adding the observed v data points before t_0 and v data points after t_1 of each WT to construct the final datasets. Note that $1 < t_0 - v < t_1 + v < T$. Hence, we have the extended dataset $\mathbf{X}_e^p = [X_e^{1,p}, \ldots, X_e^{n-1,p}, X_e^{n+1,p}, \ldots, X_e^{N,p}]^T \in \mathbb{R}^{(N-1)\times(t_1-t_0+2v+1)}$ and the extended missing dataset $Y_e^{n,p} \in \mathbb{R}^{1\times(t_1-t_0+2v+1)}$ for model training.

The extended data points of j_{th} WT are first standardized [21]:

$$\widehat{X}_e^{j,p} = \frac{X_e^{j,p} - \overline{X_e^{j,p}}}{\sigma_j} \text{ if } j = 1, \ldots, n-1, n+1, \ldots, N, \tag{3}$$

$$\widehat{Y}_e^{j,p} = \frac{Y_e^{j,p} - \overline{Y_e^{j,p}}}{\sigma_j} \text{ if } j = n, \tag{4}$$

where $\overline{(\bullet)}$ represents the mean value of the extended data and σ_j is the standard deviation of j_{th} WT.

However, it is worth noticing that, not the signals of all $N-1$ WTs are highly correlated to the missing signals of the failure WT, as shown in Fig. 3(a). We need to choose the most relevant WTs for imputation. After standardization, we compute the cross-correlation coefficient [22] matrix

$$\mathbf{C}^p = \left[C^{1,p}, \ldots, C^{n-1,p}, C^{n+1,p}, \ldots, C^{N,p}\right]^T \in \mathbb{R}^{(N-1)\times(2l+1)} \tag{5}$$

according to standardized datasets $\widehat{\mathbf{X}}_e^p$ and $\widehat{Y}_e^{n,p}$, where:

$$C^{j,p} = \mathbb{E}\left(\widehat{Y}_{e,t}^{n,p}\widehat{X}_{e,t+\tau}^{j,p}\right) \ \forall j = 1, \ldots, n-1, n+1, \ldots, N, \tag{6}$$

l is the max number of shift, $\tau = -l, \ldots, 0, \ldots, l$ is the shift number, and j is the WT number. Since noise effects will cause small increases in cross-correlation coefficients, cross-correlation vector $C^{j,p} = \{c_\tau^{j,p}\}_{\tau=-l}^{l} \in \mathbf{C}^p$ of j_{th} WT should be subtracted by introducing a penalty term $\rho(\tau)$ associated with its shift number τ, such that:

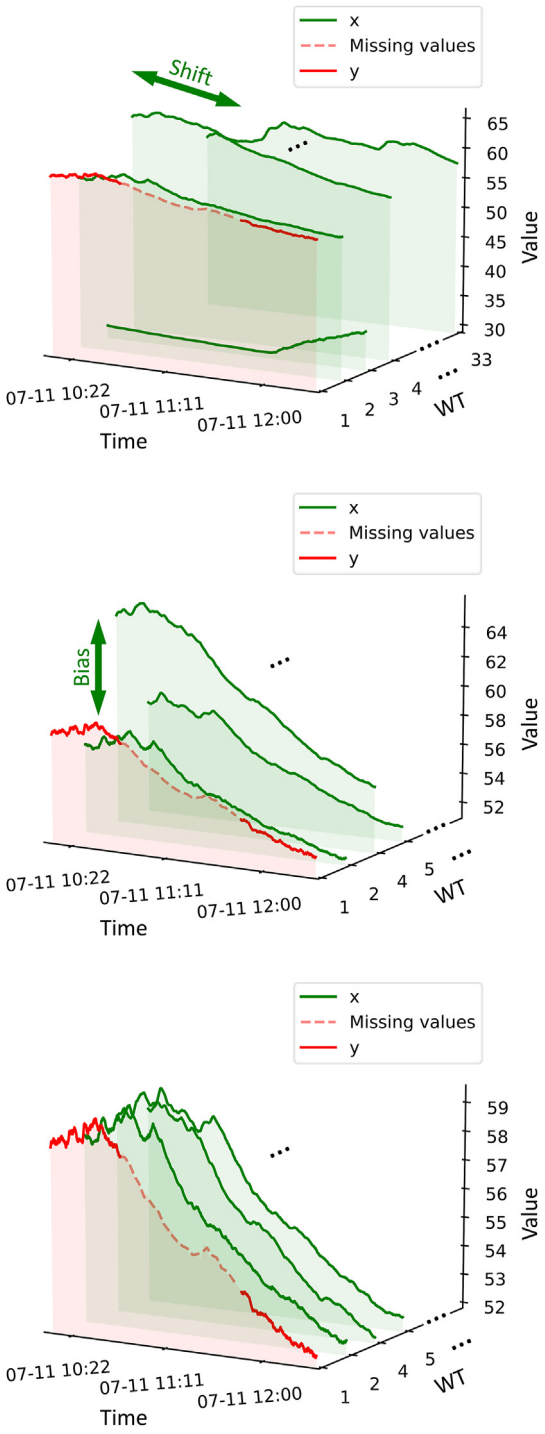


Fig. 3. Imputation process of Lazy ST. (a) A original feature of all WTs. (b) The feature after selecting and shifting. (c) The feature after offsetting bias.

$$\widehat{C}^{j,p} = \left\{c_\tau^{j,p} - \rho(\tau)\right\}_{\tau=-l}^{l} \text{ for } j = 1, \ldots, n-1, n+1, \ldots, N, \tag{7}$$

and

$$\rho(\tau) = \min(\omega \times |\tau|,\ \rho_{max}) \text{ for } \tau = -l, \ldots, 0, \ldots, l, \tag{8}$$

where ω is the weight of shift number and ρ_{max} is the pre-defined maximum penalty value. The correlation coefficient is defined as

the maximum value among the coefficient vector of each WT, i.e., $c_{max}^{j,p} = \max_{\tau=-l,\dots,l} \widehat{C}^{j,p}$ for all $j = 1,\dots,n-1,n+1,\dots,N$. Finally, the correlation coefficients of $N-1$ WTs are $\widehat{C}_{max}^{p} = \{c_{max}^{1,p},\dots,c_{max}^{n-1,p}, c_{max}^{n+1,p},\dots,c_{max}^{N,p}\}$. We keep the largest K values from $\widehat{C}_{max}^{p}$ that are most correlated with the failure WT, i.e, $\{c_{max}^{j1,p},c_{max}^{j2,p},\dots,c_{max}^{jK,p}\} \subseteq \widehat{C}_{max}^{p}$. The corresponding observed dataset after aligning is now changed to $\tilde{\mathbf{X}}_e^p = [X_e^{j1,p}, X_e^{j2,p},\dots,X_e^{jK,p}]^T \in \mathbb{R}^{K\times(t_1-t_0+2v+1)}$.

However, there exists differences between each element in $\tilde{\mathbf{X}}_e^p$ and the missing feature $\widehat{Y}_e^{n,p}$, as seen in Fig. 3(b), due to the wind velocity deficit in wake regions. Here, we introduce a bias term to compensate the differences, where the observed dataset after bias compensation is as Fig. 3(c), and the formula is:

$$\tilde{\mathbf{X}}_b^p = \tilde{\mathbf{X}}_e^p - \mathbf{b}^p, \tag{9}$$

where $\mathbf{b}^p$ is the bias factor, and $b^{jk,p}$ is calculated by linearly interpolating values [23] at both ends of the missing period:

$$\mathbf{b}^p = \left[b^{j1,p}, b^{j2,p},\dots,b^{jK,p}\right]^T \in \mathbb{R}^{K\times(t_1-t_0+2v+1)}. \tag{10}$$

After bias compensation, we further select WTs by calculating the MAE [24] from K remaining WTs. MAE value of each WT is calculated through extended segments of the data at both ends. In the end, S number of WTs with the smallest MAEs are used. The final observed dataset for imputation is $\tilde{\mathbf{X}}_{b,s}^p = [X_b^{j(1),p}, X_b^{j(2),p},\dots,X_b^{j(S),p}]^T \in \mathbb{R}^{S\times(t_1-t_0+2v+1)}$, where $\{j(1), j(2), \dots, j(S)\} \subseteq \{j1,j2,\dots,jK\}$. For each selected WT, $X_b^{j(s),p} = \{x_{b,i}^{j(s),p}\}_{i=t_0-v}^{t_1+v}$.

The imputation values $Y^{n,p}$ of the n_{th} WT in Eq. (1) are the average value of the final S WTs after bias compensation from t_0 to t_1:

$$Y^{n,p} = \frac{1}{S}\sum_s \left\{x_{b,i}^{j(s),p}\right\}_{i=t_0}^{t_1}. \tag{11}$$

Note that Eqs. (1)–(11) are developed for the imputation of the p_{th} feature. This process should be repeated P times to impute all missing features for the missing condition A in Fig. 1.

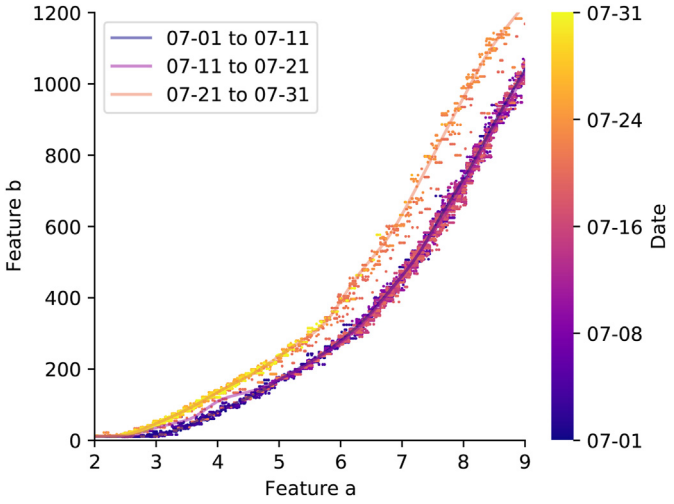


Fig. 4. Correlations between two features (e.g., a and b) on different days. Dotted points represent the actual correlation values, and the solid lines are fitted lines of the dot points. Clearly, the data pattern changes over time.

data that has acquired data for several months. In addition, from a data-driven perspective, this algorithm can deal with a degrading mechanical system (e.g., different feature correlations are observed with the degradation of a specific WT system, as shown in Fig. 4) by proposing online learning to dynamically adapt to new patterns in the data.

2.2.2.1. Learning stage. Define the dataset to be imputed, $Y^{n,p}$, of the p_{th} feature of the n_{th} WT as a sequence of time-series data as:

$$Y^{n,p} = \left\{x_i^{n,p}\right\}_{i=1}^{T} \in \mathbb{R}^{1\times T}, \tag{12}$$

where $p = 1,2,\dots,P$, $n \le N$. The rest, $P-1$ features, construct the observed dataset, which can be defined as:

$$\mathbf{X}^n = \left[X^{n,1},\dots,X^{n,p-1},X^{n,p+1},\dots,X^{n,P}\right]^T \in \mathbb{R}^{(P-1)\times T}, \tag{13}$$

where each element of X^n represents the observed data points of that feature. Since using all the features will slow down the training and inference process, only a few of them have a strong correlation with $Y^{n,p}$; therefore, $\mathbf{X}^n$ needs to be selected.

Algorithm 1
Lazy ST

Input: The extended time series $Y_e^{n,p}$ of the p_{th} feature of the n_{th} WT. The matrix of other WTs data, $\mathbf{X}_e^p$.
Output: Data after imputation, $Y^{n,p}$.
Standardize the elements in $\mathbf{X}_e^p$, $Y_e^{n,p}$ as Equations (3) and (4) and get $\tilde{\mathbf{X}}_e^p$, $\tilde{Y}_e^{n,p}$.
Compute the cross-correlation coefficient matrix $\mathbf{C}^p$ as Equation (6).
Add the shift-penalty term to $\mathbf{C}^p$ as Equation (7) and get the matrix $\widehat{\mathbf{C}}^p$.
Select K WTs with the highest correlation coefficient in $\widehat{\mathbf{C}}^p$ after aligning and get the matrix $\tilde{\mathbf{X}}_e^p$.
Compensate the bias as Equation (9) and get $\tilde{\mathbf{X}}_b^p$.
Select S WTs with the smallest MAE and get $\tilde{\mathbf{X}}_{b,s}^p$.
Average the selected WT data $\tilde{\mathbf{X}}_{b,s}^p$ as Equation (11) and get the result $Y^{n,p}$.
return $Y^{n,p}$;

2.2.2. Online variation algorithm for time-varying system

In terms of the missing-data condition (b) in Fig. 1, we propose an online-learning algorithm that imputes missing values using only their neighboring points with strong fitting ability. This algorithm is especially suited for large-scale datasets, such as wind farm

We use the XGBoost algorithm [25] to fit the data pattern between $Y^{n,p}$ and $\mathbf{X}^n$, which generates feature importance during the fitting process. The most important feature is selected as input:

$$X^{n,q} \in \mathbb{R}^{1\times T},\ q \in \{1,\ldots,p-1,p+1,\ldots,P\}, \tag{14}$$

which has the highest importance score.

The $X^{n,q} = \{X^{n,q,\varphi}\}_{\varphi=1}^{\Phi}$ is segmented to Φ parts according to the distribution range and then focused on the locality:

$$X^{n,q,\varphi} = \left\{x_i^{n,q} | r_{\varphi-1} < x_i^{n,q} \le r_\varphi\right\}_{i=t_2}^{t_3}, \tag{15}$$

and

$$r_\varphi = \varphi \cdot \frac{x_{max}^{n,q} - x_{min}^{n,q}}{\Phi} + x_{min}^{n,q}, \tag{16}$$

where $\varphi = 1,\ldots,\Phi$ is the segment number, $x_{max}^{n,q} = \max_{i=1,\ldots,T} X^{n,q}$ is the maximum of the $X^{n,q}$ and $x_{min}^{n,q} = \min_{i=1,\ldots,T} X^{n,q}$ is the minimum of the $X^{n,q}$. As the aging of equipment or the change of environment is not too serious in a short period of time, we set up a time window to model every $t_3 - t_2$ days. Thus, the $Y^{n,p,\varphi}$ can be defined accordingly:

$$Y^{n,p,\varphi} = \left\{y_i^{n,p} | x_i^{n,q} \in X^{n,q,\varphi}\right\}_{i=t_2}^{t_3}. \tag{17}$$

For each segment, we use the median value to eliminate the volatility impact of the maximum and minimum values on the results, such that:

$$\begin{aligned} X_{med}^{n,q} &= \left\{x_{med}^{n,q,\varphi}\right\}_{\varphi=1}^{\Phi}, \\ Y_{med}^{n,p} &= \left\{y_{med}^{n,p,\varphi}\right\}_{\varphi=1}^{\Phi}, \end{aligned} \tag{18}$$

where $x_{med}^{n,q,\varphi} = \text{median}(X^{n,q,\varphi})$ is the median of the $X^{n,q,\varphi}$ and $y_{med}^{n,p,\varphi} = \text{median}(Y^{n,p,\varphi})$ is the median of the $Y^{n,p,\varphi}$.

Additionally, the coefficient of quartile variation (CQV) [26,27] is used to describe the statistical dispersion in this case as follows:

$$cqv = \frac{Q_3 - Q_1}{Q_3 + Q_1}, \tag{19}$$

where Q_1 and Q_3 are the first and third quartiles of each segment, respectively. Additionally, we detect and remove the outliers by the 3σ rule [28] to enhance the robustness of our algorithm.

2.2.2.2. Inference stage. When we want to impute the missing value $y_i^{n,p}$, we search the $x_i^{n,q}$, which has the same time stamp *i*. Then we can interpolate it with the Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) [29,30] method to get the corresponding $y_i^{n,p}$:

$$y_i^{n,p} = f_{\text{PCHIP}}\left(X_{med}^{n,q}, Y_{med}^{n,p}, x_i^{n,q}\right). \tag{20}$$

Compared to a linear interpolation method, PCHIP generates smoother results and fits better for nonlinear cases. Moreover, it is more stable than other interpolation methods, such as spline [30]. At the same time, the corresponding dispersion coefficient is returned to determine the confidence of the result. This is similar to kernel regression [31] in that both are non-parametric and local-fitting algorithms. However, kernel regression has to traverse all the data every time it makes a prediction, which takes more time to impute missing data compared to our method. The complete algorithm of the Online Variation imputation process is shown in Algorithm 2.

Algorithm 2
Online Variation

Input: The time series $Y^{n,p}$ of the p_{th} WT of the n_{th} feature. The matrix of other features data $\mathbf{X}^n$.
Output: Data after imputation, $y_i^{n,p}$.
Select the time series $X^{n,q}$, which has the most important feature, through XGBoost algorithm.
Use the 3σ rule to filter outliers in $X^{n,q}$.
Segment the $X^{n,q}$ and $Y^{n,p}$ in the time window into Φ parts as Equation 15–17, and each part is $X^{n,q,\varphi}$ and $X^{n,p,\varphi}$.
Form median vector $X_{med}^{n,q}$ and $Y_{med}^{n,p}$ as Equation (18).
Use the PHCIP algorithm to interpolate the missing values as Equation (20) and get $y_i^{n,p}$.
return $y_i^{n,p}$;

3. Experimental verification

3.1. Dataset description

The experimental data, which was exported from an offshore wind farm through a SCADA system, is provided by the GOLDWIND corporation from a data-mining competition.[1] The sampling time of sensors is approximate to 7.4 s. This dataset contains 33 WTs with 68 features. Features gathered from different sensors include wind speed, power production, gear temperature, wind direction, blade pitch angle, etc. To simulate the missing data for imputation during system failure, this study randomly removes 26 periods of different features in the time domain. Durations of missing periods last from 6 min to 8 h to simulate short-to long-term missing proportions. And, we generate two missing-data modes as depicted in Fig. 5. The extend number *v* before and after the missing period is set to 1500. The maximum shift number is $l = 30$. The predefined maximum penalty value ρ_{max} is set to 0.05.

Our experiments performed on a PC were configured with Intel Core i5-9400F CPU, without calling multiple processes. Not all features had the spatio-temporal or feature correlation. Some features were more suitable for mode or median imputation. For the Lazy ST method, some features had linear correlation; therefore, they were highly similar, and we could impute just one of them.

3.2. Evaluation metrics

Three metrics to assess imputation performance of the proposed Lazy ST and Online Variation algorithms are employed, namely MAE, MSE, and R-squared (R^2) [24,32]. Their expressions are given as follows:

[1] DataFountain, https://www.datafountain.cn/competitions/333; 2019 [accessed 7 March 2021].

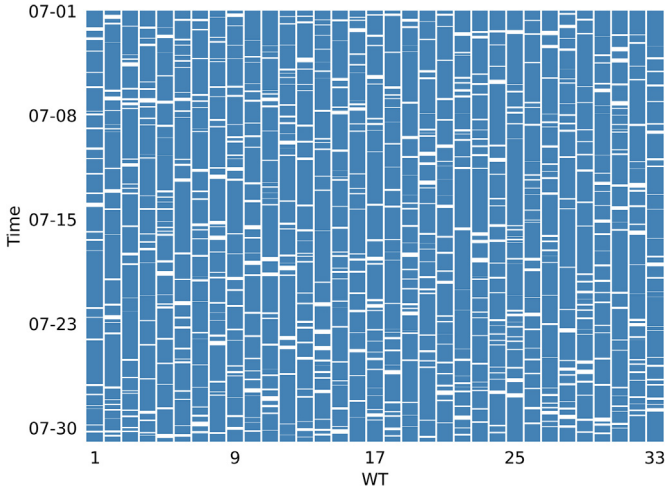


Fig. 5. Missing-data distribution of feature 1. Other features have similar distributions. The white gaps represent the missing values. The bigger the gap, the more missing values exist.

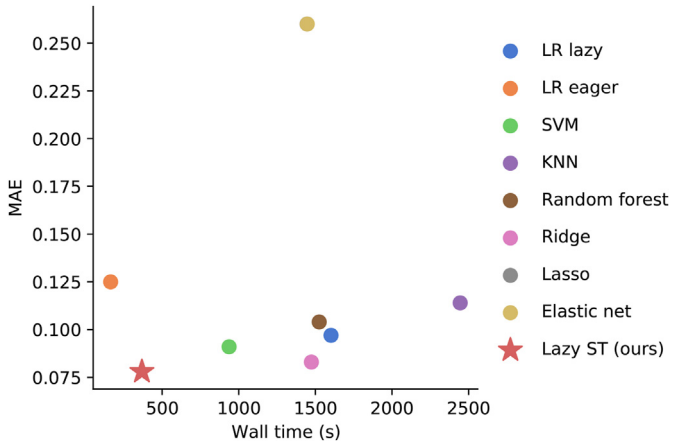


Fig. 6. MAE-time scatter figure. We use MAE value and Wall time as indicators to reflect the effectiveness and efficiency of these eight methods. Our method is marked with the red star, which is located in the lower left corner of the graph. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

$$MAE = \frac{1}{n}\sum_{t=1}^{n}\left|y_t - \widehat{y}_t\right|$$

$$MSE = \frac{1}{n}\sum_{t=1}^{n}\left(y_t - \widehat{y}_t\right)^2 \tag{21}$$

$$R^2 = 1 - \frac{\sum_{t=1}^{n}\left(y_t - \widehat{y}_t\right)^2}{\sum_{t=1}^{n}\left(y_t - \overline{y}\right)^2}$$

where $\widehat{y}_t$ and y_t are the predicted value and ground truth of the t_{th} sample, respectively.

3.3. Results and comparison of lazy ST algorithm

The performance of the Lazy ST algorithm was examined extensively by generating 26 random missing periods from each WT, ranging from 6 min to 8 h. Moreover, to verify the generalizability of our method, 11 features were selected for experimental validation, rather than just using wind speed or generated energy as in most conventional experiments.

To verify the effectiveness of the proposed algorithm, nine baseline and machine-learning methods were used for comparison purposes. Mean value was calculated for data imputation as the baseline method. Linear models, such as linear regression (LR) and linear support vector machine (SVM) algorithms, were adopted after data normalization. The input data is concatenated after shifting. We also compared the proposed methods with nonlinear models, such as k-nearest neighbors (kNN) and random forest (RF) algorithms. Sparse models such as Ridge, least absolute shrinkage and selection operator (LASSO) and elastic net were also employed for feature selection.

Table 1 shows the comparison results (mean value of 26 random missing periods) between the Lazy ST and other methods, where Wall time reflects the time consumption of each method. It can be observed that our method outperforms other methods by obtaining the lowest MAE and MSE values, highest R^2, as well as very short Wall time (<400). Fig. 6 is the MAE-time scatter plot that corresponds to the results in Table 1, indicating that our Lazy ST method results in the smallest MAE with a relatively short Wall time. The proposed method also shows the superiority of time cost, which is smaller than other methods, except for the LR eager. The results below indicate that Lazy ST has the highest accuracy and also achieves optimal trade-off between effectiveness and efficiency. Moreover, the time evolution of the imputation results of six methods are shown in Fig. 7. It can be seen that the imputation data of the Lazy ST method are almost close to the ground truth from beginning to end as compared to other methods, further demonstrating the superior performance of our method.

Table 1
Comparison of different methods. The order refers to the number of shifted steps. In order to limit the wall time around 1500 s for better comparison, the shift order is set to 5 when using LR Lazy, RF, Ridge, Lasso and Elastic net. For these methods that spend longer time or data cannot, the order is set to 0. The normalize option means whether original data will be normalized to the range [0,1].

Method	Order	Normalize	Lazy or Eager	Wall time	MAE	MSE	$R^2\uparrow$
Mean	0	✗	Eager	**38**	0.735	0.938	−0.063
LR	5	✓	Lazy	1602	0.097	0.135	0.838
LR	0	✓	Eager	164	0.125	0.107	0.808
SVM	0	✓	Lazy	937	0.091	0.094	0.887
kNN	0	✗	Lazy	2445	0.114	0.086	0.884
Random forest	5	✗	Lazy	1525	0.104	0.057	0.931
Ridge	5	✓	Lazy	1474	0.083	0.054	0.945
LASSO	5	✓	Lazy	1446	0.260	0.185	0.704
Elastic net	5	✓	Lazy	1447	0.260	0.185	0.704
Lazy ST (ours)	30	✗	Lazy	368	**0.078**	**0.048**	**0.949**

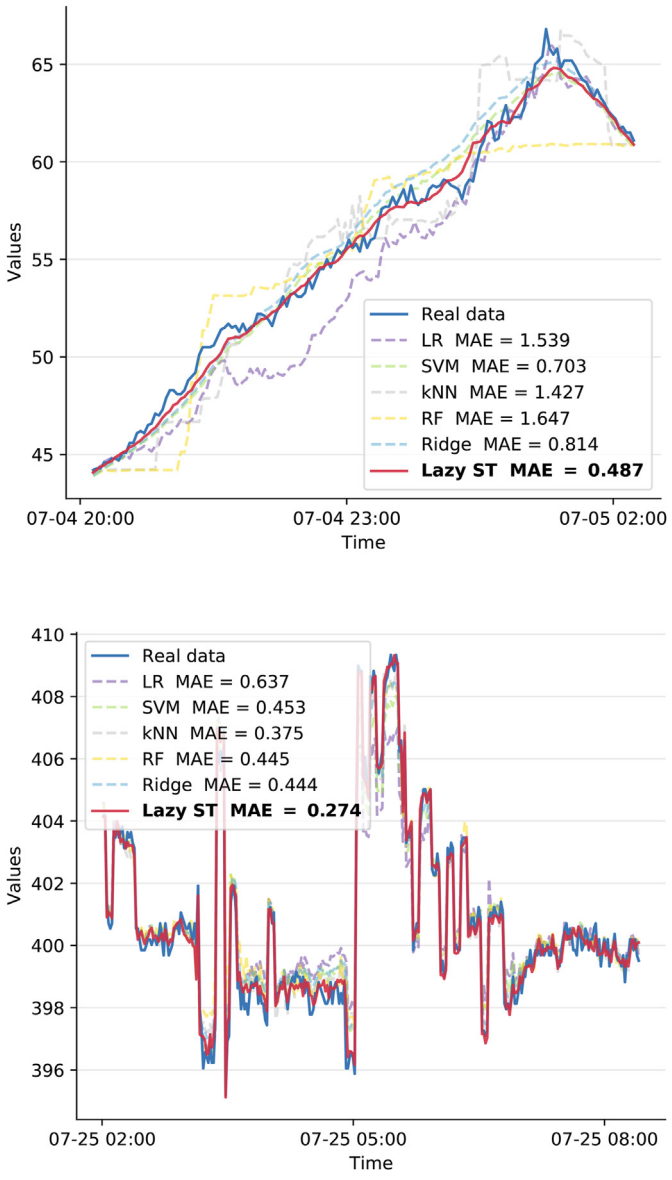


Fig. 7. Comparison of imputation results of (a) the 3_{rd} feature of the 20_{th} WT (a slow varying data case) and (b) the 1_{st} feature of the 3_{rd} WT (a fast varying data case). The blue solid line is the ground truth data, and the red solid line indicates the values predicted by our method. Other dotted lines are the imputation data of other comparison methods. It can be seen that our method performs best as it almost overlapping with the ground truth. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3.4. Lazy ST algorithm ablation study

Ablation experiments were carried out to investigate the contribution of each module (e.g., the spatio-temporal correlation module, penalty module, compensate bias module, as well as MAE selection module described in Section 2.2.1) of the algorithm. The original Lazy ST algorithm, without any modification, is compared to the four tested modules, which are 1) The spatio-temporal correlation module, used for minimizing phase differences, is eliminated from the original algorithm (denoted by "w/o ST correlation"); 2) The penalty module, used to constrain unrestricted shifts, is eliminated from the original algorithm (denoted by "w/o penalty"); 3) The offset bias module, used for offsetting the bias between WTs, is eliminated from the original algorithm (denoted by "w/o offset bias"); and 4) The MAE selection module, used to select the most similar WTs, is eliminated from the original algorithm (denoted by "w/o MAE selection").

Table 2
Ablation comparison between the proposed Lazy ST and the other nine conventional methods.

Deleted module	MAE	MSE	$R^2 \uparrow$
w/o ST correlation	0.0795	0.0494	0.9466
w/o penalty	0.0782	0.0477	0.9491
w/o offset bias	0.1497	0.1166	0.8590
w/o MAE selection	0.0797	0.0478	0.9489
Lazy ST	**0.0781**	**0.0477**	**0.9492**

As shown in Table 2, the proposed Lazy ST without any observed modification has the best imputation performance, as it obtains the smallest MAE and MSE values and the largest R^2. Thus, each module of the algorithm has a degree of contribution to the accuracy of the imputed data.

In Fig. 8, the coefficient of cross-correlation between WTs is more vividly visualized. Different polar angles and radial coordinates represent different WT numbers and shifts, respectively. When the radial coordinate is greater than 0, it represents a positive shift in time domain. However, coordinates less than 0 refer to a negative shift. Different shades of color represent different values of Pearson correlation coefficient; the dark red represents a high correlation between two WTs, while the dark blue indicates there is no correlation. It can be observed that most of the WTs reach a high degree of correlation after shifting.

3.5. Results and comparison of online variation algorithm

The validity of the algorithm is proved by selecting three pairs of features. More than 26 consecutive periods of data have been randomly deleted. In addition, the data of four specific numerical intervals were deleted to test the algorithm's inferential ability. Similar to the Lazy ST algorithm, mean value is calculated as the baseline method. The kNN and the kernel regression (KR) algorithms are also compared, as they are non-parameter methods for nonlinear data imputation like our algorithm.

Comparison results (mean value of 26 random missing periods)

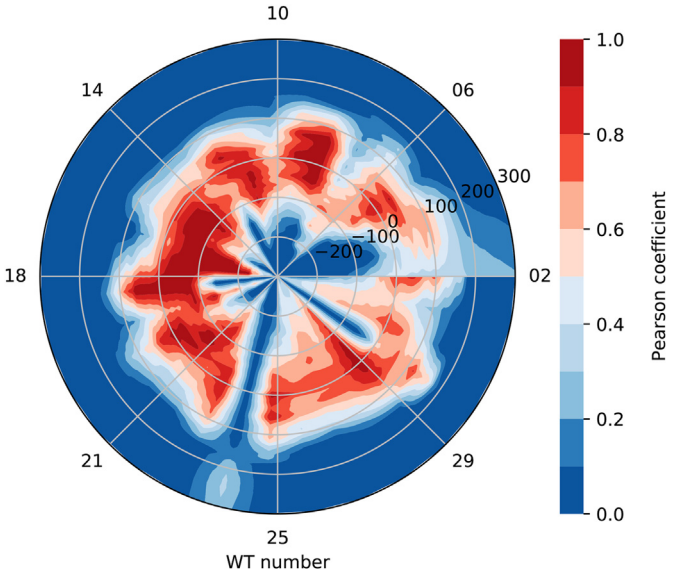


Fig. 8. Cross-correlation of the 2_{nd} feature of the 1_{st} WT.

Table 3
Comparisons of different methods of the Online Variation algorithm.

Method	Wall time	MAE	MSE	$R^2 \uparrow$
Mean	**11**	0.736	0.8266	0.0387
kNN	54	0.055	0.0079	0.9904
Online kNN	38	0.053	0.0073	0.9910
KR	522	0.029	0.0022	0.9974
Online KR	706	0.026	0.0021	0.9976
Online Variation (ours)	20	**0.023**	**0.0014**	**0.9984**

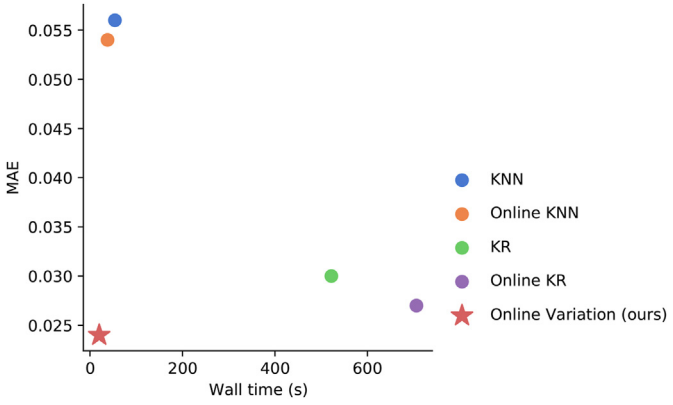


Fig. 9. MAE-time scatter figure.

in Table 3 show that the proposed Online-Variation method achieves the best results, as it significantly reduces the MAE and MSE values and increases the R^2 compared to other methods. The all-time result of our Online-Variation method is only slightly worse than the baseline mean method, but it is much faster than the other comparison methods. To further investigate the performance, MAE-time scatter figure is presented in Fig. 9. In this case, our method is found to be the best method for data imputation in terms of time consumption and imputation precision. In order to evaluate the online learning performed by Eq. (15), the fusion of online learning with kNN and KR algorithms was also tested (denoted by "Online kNN" and "Online KR" in Table 9, respectively). The performance increases, i.e., the decrease of MAE and MSE, as well as the increase of R^2, confirm the effectiveness of the online-learning module used in this study. Additionally, the imputation results of five methods of the 34_{th} feature of the 8_{th} WT are shown in Fig. 10. It is obvious that our method generates the best result, which almost overlaps with the ground truth, while the sharp points of other methods exceed the ground truth. This further proves the effectiveness of our method.

3.6. Influence of signal noise

Considering that sensor noise may cause feature errors, we added specific amplitude Gaussian noises to evaluate the impact of feature uncertainties in the proposed Lazy ST and Online Variation algorithms. More specifically, the amplitudes of noise are set to 1%, 3%, and 5% standard deviation of each feature. Results in Tables 4 and 5 show that even though the MAE values increase slightly as the amplitude increases, our methods still achieved the smallest MAEs under all the noise conditions, which demonstrates that our methods can resist a certain level of feature error affected by sensor noises.

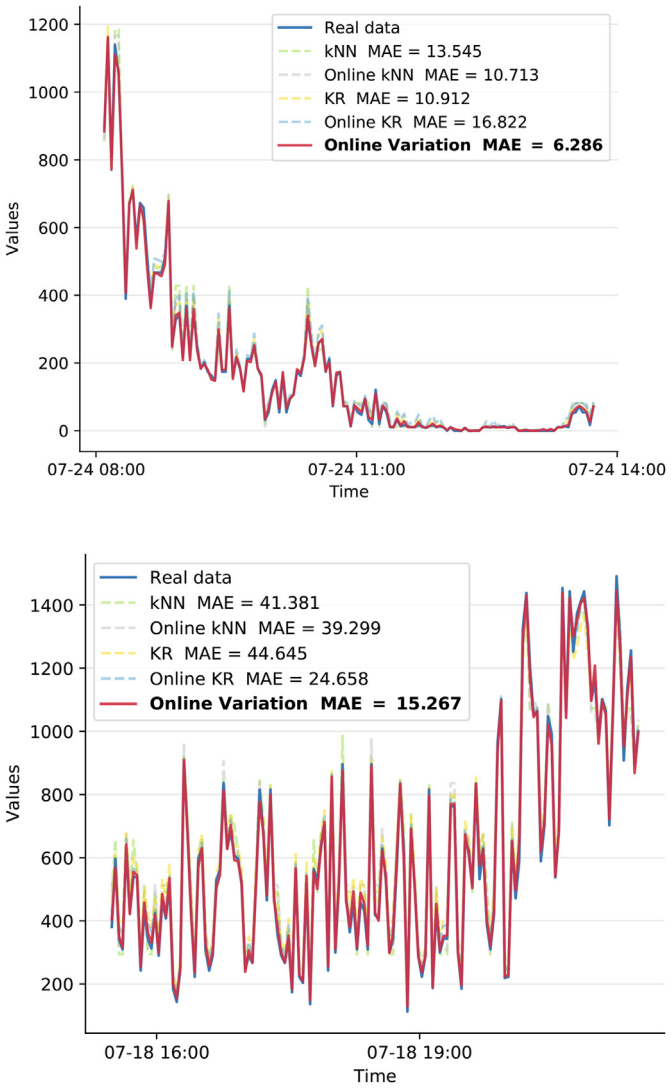


Fig. 10. Comparison of imputation results of (a) the 34_{th} feature of the 8_{th} WT (a slow varying data case) and (b) the 34_{th} feature of the 14_{th} WT (a fast varying data case). The blue solid line is the ground truth data, and the red solid line indicates the values predicted by our method. Other dotted lines are the imputation data of other comparison methods. Obviously, our method almost overlaps with the ground truth and achieves the lowest MAE. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 4
Comparison of different methods for Lazy ST with Gaussian noise.

Method	MAE			
	0%	1%	3%	5%
Mean	0.735	0.735	0.735	0.735
LR	0.097	0.095	0.094	0.095
LR Eager	0.125	0.125	0.125	0.125
SVM	0.091	0.090	0.089	0.090
kNN	0.114	0.114	0.116	0.119
Random forest	0.104	0.105	0.106	0.109
Ridge	0.083	0.083	0.084	0.084
Lasso	0.260	0.260	0.260	0.260
Elastic net	0.260	0.260	0.260	0.260
Lazy ST (ours)	**0.078**	**0.079**	**0.080**	**0.082**

Table 5
Comparison of different methods for Online Variation with Gaussian noise.

Method	MAE			
	0%	1%	3%	5%
Mean	0.736	0.733	0.729	0.722
kNN	0.055	0.056	0.064	0.077
Online kNN	0.053	0.054	0.063	0.076
KR	0.029	0.031	0.040	0.054
Online KR	0.026	0.030	0.038	0.052
Online Variation (ours)	**0.023**	**0.025**	**0.035**	**0.048**

Table 6
Comparison of different methods using Lazy ST with different number of WTs.

Method	MAE					
	5 WTs	10 WTs	15 WTs	20 WTs	25 WTs	33 WTs
Mean	0.732	0.746	0.744	0.725	0.735	0.735
LR	0.121	0.105	0.093	0.092	0.100	0.097
LR Eager	0.192	0.139	0.127	0.126	0.131	0.125
SVM	**0.114**	0.100	0.088	0.088	0.093	0.091
kNN	0.171	0.123	0.114	0.110	0.115	0.114
Random forest	0.132	0.116	0.107	0.103	0.108	0.104
Ridge	0.126	0.101	0.088	0.083	0.088	0.083
LASSO	0.273	0.270	0.259	0.252	0.263	0.260
Elastic net	0.274	0.270	0.259	0.252	0.263	0.260
Lazy ST (ours)	0.119	**0.095**	**0.081**	**0.077**	**0.081**	**0.078**

3.7. Influence of wind farm configuration

Since different wind farm configurations can affect the performance of imputation methods, we conducted a series of experiments to verify the robustness and generalizability of our methods. To this end, we carried out the following experiments with respect to different sizes of WTs, different sampling frequencies, and different locations of offshore wind farms.

Different sizes of WTs on a wind farm: We evenly selected 5, 10, 15, 20, and 25 of the 33 WTs. The imputation results of the different sizes using the Lazy ST algorithm are in Table 6. It can be seen that our method obtained the smallest MAEs when the wind farm had more than 5 WTs (and only 0.005 larger than the SVM method in the case of 5 WTs). The Online Variation method was not tested, as it imputes missing values in the same WT, which is not relevant for this experimental scenario.

Different sampling frequencies in a wind farm: In addition to the original sampling interval (i.e., 7.4 s) of the SCADA system, we also reconstructed the dataset with 1-, 2-, and 3-min time intervals. As shown in Tables 7 and 8, the performance of our methods degraded as the time interval increased, but it is still superior to other methods.

Table 7
Comparison of different methods using Lazy ST with different sampling intervals.

Method	MAE			
	7.4 s	1 min	2 min	3 min
Mean	0.735	0.736	0.738	0.738
LR	0.097	0.380	0.125	0.107
LR Eager	0.125	0.126	0.126	0.126
SVM	0.091	0.090	0.089	0.091
kNN	0.114	0.115	0.117	0.118
Random forest	0.104	0.103	0.105	0.107
Ridge	0.083	0.084	0.086	0.087
LASSO	0.260	0.263	0.265	0.264
Elastic net	0.260	0.263	0.263	0.262
Lazy ST (ours)	**0.078**	**0.080**	**0.081**	**0.082**

Table 8
Comparison of different methods using Online Variation with different sampling intervals.

Method	MAE			
	7.4 s	1 min	2 min	3 min
Mean	0.736	0.743	0.744	0.746
kNN	0.055	0.055	0.055	0.054
Online kNN	0.053	0.052	0.053	0.053
KR	0.029	0.030	0.029	0.030
Online KR	0.026	0.027	0.028	0.027
Online Variation (ours)	**0.023**	**0.023**	**0.023**	**0.024**

Table 9
Comparison of different methods using Lazy ST on UEBB and UEPS datasets.

Method	MAE	
	UEBB	UEPS
Mean	0.771	0.814
LR	0.286	0.144
LR Eager	0.162	0.157
SVM	0.158	0.097
kNN	0.137	0.108
Random forest	0.110	0.078
Ridge	0.126	0.094
LASSO	0.351	0.277
Elastic net	0.405	0.333
Lazy ST (ours)	**0.105**	**0.073**

Additional offshore wind farm datasets from Brazil: For this scenario, we conducted experiments on two additional open-source datasets [33] that have four features: active power, rotor angular speed, nacelle wind speed, and act position turbine variable. These two wind farms are on the northeast coast of Brazil: Beberibe Wind Farm (UEBB) and Pedra do Sal Wind Farm (UEPS). UEBB and UEPS have 32 and 20 WTs, respectively.

The datasets from their SCADA systems were both collected between August 2013 to July 2014 at a 10-min interval. All four features were used to evaluate the performance of Lazy ST, and the active power was used to evaluate the performance of Online Variation. The missing data of these two additional datasets were generated similarly to the competition dataset. Twenty-six periods of different features in time domain were randomly removed, and the duration of missing periods last from 20 min to 8 h. The promising results in Tables 9 and 10 further confirm the generalizability of our methods.

4. Conclusion

In this paper, we proposed a learning framework to simultaneously impute two missing-data conditions. First, a lazy spatio-temporal correlation algorithm was proposed to impute the situation that all features of a WT are missing. In the case that one specific feature of all wind turbines is missing, an Online-Variation

Table 10
Comparison of different methods using Online Variation on UEBB and UEPS datasets.

Method	MAE	
	UEBB	UEPS
Mean	0.831	0.838
kNN	0.137	0.156
Online kNN	0.116	0.124
KR	0.095	0.121
Online KR	0.080	0.081
Online Variation (ours)	**0.072**	**0.073**

learning algorithm based on inter-features correlation was proposed to fit the data pattern, which is suitable for the aging of a time-varying system over time. Experiments, based on an offshore wind farm dataset consisting of 33 wind turbines with 68 features, were conducted to demonstrate the effectiveness of the two proposed algorithms for two missing conditions, respectively. Compared to other traditional imputation methods, lower MAE and MSE values, and higher R^2 with shorter computation time was observed for the two proposed algorithms for different missing conditions. The experiment results have verified that our online-learning methods perform superiorly to the compared methods in both speed and accuracy. Additionally, we carried out experiments with respect to different amplitudes of sensor noise, different sizes of WTs, different sampling frequencies, and different locations of offshore wind farms to demonstrate the robustness and generalizability of our methods.

Future work will include using a graph neural network (GNN) to explore space information among WTs and long short-term memory (LSTM) to fit time-series information.

Credit author statement

Sun Chuan: Conceptualization, Methodology, Software **Chen Yueyi:** Validation, Writing - Original Draft **Cheng Cheng:** Conceptualization, Writing - Review & Editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported in part by the National Natural Science Foundation of China [Grant number 51905197], and in part by the National Key R&D Program of China [Grant number 2018YFB1701202].

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